CS1231S Discrete Structures

WEEK 7 RECITATION
LECTURE 8. MATHEMATICAL INDUCTION

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CS1231S (AY2026/27 Sem1) QnA

Sequences

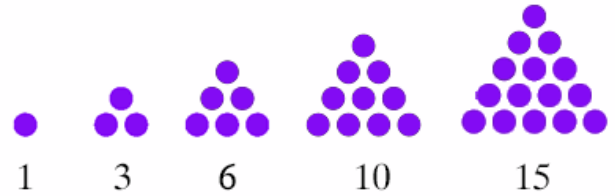
Definitions: Sequence and Terms

A **sequence** is an ordered set with members called **terms**.
Usually, the terms are numbers. A sequence may have infinite terms.

General form: $a_m, a_{m+1}, a_{m+2}, \cdots, a_n$ where $m \leq n$.
Infinite sequence: $a_m, a_{m+1}, a_{m+2}, \cdots$

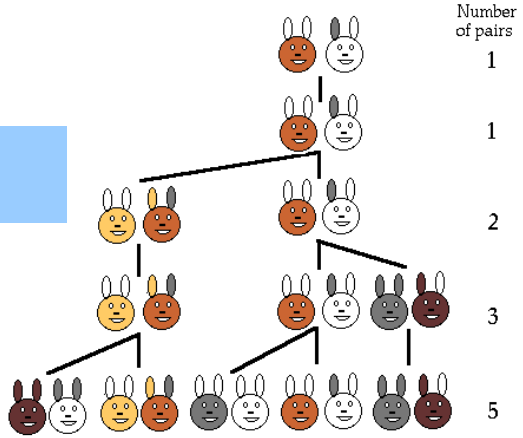
Squares: 1, 4, 9, 16, 25, 38, 49, …

Triangle numbers: 1, 3, 6, 10, 15, 21, 28, …

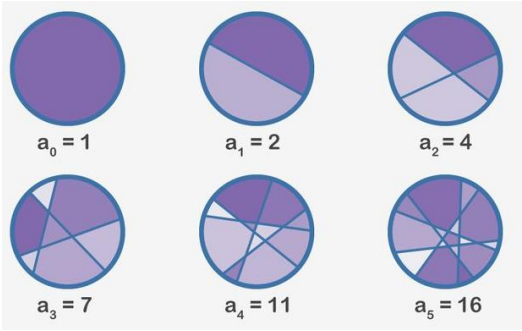


Prime numbers: 2, 3, 5, 7, 11, 13, 17, 19, …

Fibonacci numbers: 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, …



Lazy caterer's sequence: 1, 2, 4, 7, 11, 16, …
(See AY2018/19 semester 1 exam paper.)



Sequences: Summation Notation

Definition: Summation

If m and n are integers, $m \leq n$, the symbol

$$\sum_{k=m}^{n} a_k$$

is the **sum** of all the terms $a_m, a_{m+1}, a_{m+2}, \cdots, a_n$.

We say that $a_m + a_{m+1} + a_{m+2} + \cdots + a_n$ is the **expanded form** of the sum, and we write

$\sum_{k=m}^{n} a_k = a_m + a_{m+1} + a_{m+2} + \cdots + a_n$.

We call k the **index** of the summation, m the **lower limit** of the summation and n the **upper limit** of the summation.

Recursive definition:

Let m be any integer.

$$\sum_{k=m}^{m} a_k = a_m$$

$$\sum_{k=m}^{n} a_k = \left(\sum_{k=m}^{n-1} a_k \right) + a_n \text{ for all integers } n > m.$$

By convention, an empty sum (eg: $\sum_{k=m}^{n} a_k$ where $m > n$) is equal to the additive identity 0.

Sequences: Product Notation

Definition: Product

If m and n are integers, $m \leq n$, the symbol

$$\prod_{k=m}^{n} a_k$$

is the **product** of all the terms

$a_m, a_{m+1}, a_{m+2}, \cdots, a_n$.

We write

$$\prod_{k=m}^{n} a_k = a_m \cdot a_{m+1} \cdot a_{m+2} \cdot \cdots \cdot a_n.$$

Recursive definition:

Let m be any integer.

$$\prod_{k=m}^{m} a_k = a_m$$

$$\prod_{k=m}^{n} a_k = \left(\prod_{k=m}^{n-1} a_k\right) \cdot a_n \text{ for all integers } n > m.$$

By convention, an empty product (eg: $\prod_{k=m}^{n} a_k$ where $m > n$) is equal to the multiplicative identity 1.

Sequences: Properties of Summations and Products

Theorem 5.1.1

If $a_m, a_{m+1}, a_{m+2}, \cdots$ and $b_m, b_{m+1}, b_{m+2}, \cdots$ are sequences of real numbers and c is any real number , then the following equations hold for any integer $n \geq m$:

1. $$\sum_{k=m}^{n} a_k + \sum_{k=m}^{n} b_k = \sum_{k=m}^{n} (a_k + b_k)$$

2. $$c \cdot \sum_{k=m}^{n} a_k = \sum_{k=m}^{n} c \cdot a_k$$ (generalized distributive law)

3. $$\left(\prod_{k=m}^{n} a_k\right) \cdot \left(\prod_{k=m}^{n} b_k\right) = \prod_{k=m}^{n} (a_k \cdot b_k)$$

Sequences: Properties of Summations and Products

Example #6:

Let $a_k = k + 1$ and $b_k = k - 1$ for all integers k.

(a) Write the following as a single summation;

$$\sum_{k=m}^{n} a_k + 2 \cdot \sum_{k=m}^{n} b_k$$

$$= \sum_{k=m}^{n} (k+1) + 2 \cdot \sum_{k=m}^{n} (k-1) \quad \text{(by substitution)}$$

$$= \sum_{k=m}^{n} (k+1) + \sum_{k=m}^{n} 2 \cdot (k-1) \quad \text{(by Theorem 5.1.1 (2))}$$

$$= \sum_{k=m}^{n} \big((k+1) + 2 \cdot (k-1)\big) \quad \text{(by Theorem 5.1.1 (1))}$$

$$= \sum_{k=m}^{n} (3k-1) \quad \text{(by basic algebra)}$$

(b) Write the following as a single product.

$$\left(\prod_{k=m}^{n} a_k\right) \cdot \left(\prod_{k=m}^{n} b_k\right)$$

$$= \left(\prod_{k=m}^{n} (k+1)\right) \cdot \left(\prod_{k=m}^{n} (k-1)\right) \quad \text{(by substitution)}$$

$$= \prod_{k=m}^{n} (k+1) \cdot (k-1) \quad \text{(by Theorem 5.1.1 (3))}$$

$$= \prod_{k=m}^{n} (k^2 - 1) \quad \text{(by basic algebra)}$$

Sequences: Change of Variable

$$\sum_{k=1}^{3} k^2 = \sum_{i=1}^{3} i^2 = \sum_{k=3}^{5} (k-2)^2$$

Dummy variables

Example #7: Transform the following summation by changing the range of k from $[1, n+1]$ to $[0, n]$.

$$\sum_{k=1}^{n+1} \left(\frac{k}{n+k}\right) = \sum_{k=0}^{n} \left(\frac{k+1}{n+k+1}\right)$$

Mathematical Induction: The Domino Theory



Imagine an infinite row of dominoes, numbered $D_1, D_2, D_3, \cdots$

We want to prove that every domino will eventually fall.

Let $P(k)$ means domino k falls.

Strategy: We need two things:

Fact 1: Someone pushes D_1, so D_1 falls.

Fact 2: Whenever D_k falls, it knocks down D_{k+1}.

Basis step: $P(1)$

Inductive step: $P(k) \rightarrow P(k+1) \forall k \geq 1$

Therefore, all dominoes fall!

Conclusion: $P(n)$ for all integers $n \geq 1$

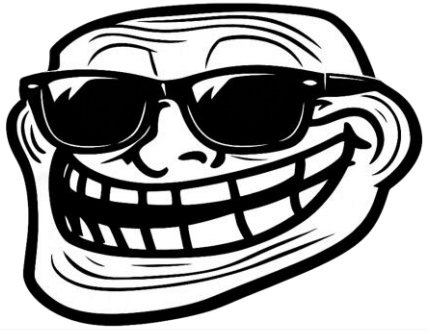
Mathematical Induction

Basis step: $P(1)$

Inductive step: $P(k) \rightarrow P(k+1)\ \forall k \geq 1$

Conclusion: $P(n)$ for all integers $n \geq 1$

Your job is to prove these 2 statements



Mathematical induction is How to Knock Down Infinitely Many Dominoes with only 2 proofs!

Unrolling the argument:

1. $P(1)$ is true! (by the basis step)
2. $P(2)$ is also true! (by modus ponens on $P(1)$ and $P(1) \rightarrow P(2)$.)
3. $P(3)$ is also true! (by modus ponens on $P(2)$ and $P(2) \rightarrow P(3)$.)
4. $P(4)$ is also true! (by modus ponens on $P(3)$ and $P(3) \rightarrow P(4)$.)
:

Mathematical Induction

Principle of Mathematical Induction (PMI)

Let $P(n)$ be a property that is defined for integers n, and let a be a fixed integer. Suppose the following 2 statements are true:

1. $P(a)$ is true.
2. For all integers $k \geq a$, if $P(k)$ is true then $P(k+1)$ is true.

Then the statement "for all integers $n \geq a, P(n)$" is true.

Inductive step: For all integers $k \geq a$, $P(k) \rightarrow P(k+1)$

This is called the inductive hypothesis.

Mathematical Induction

Example #8:

Use mathematical induction to prove:

Theorem 5.2.2 (5th: 5.2.1) Sum of the First n Integers

For all integers $n \geq 1$,

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

Proof (by *mathematical induction*):

1. Let $P(n) \equiv \left(1 + 2 + \cdots + n = \frac{n(n+1)}{2}\right), \forall n \in \mathbb{Z}^+$. (Set up predicate.)
2. Basis step: $1 = \frac{1(1+1)}{2}$, therefore $P(1)$ is true.
3. Assume $P(k)$ is true for arbitrary $k \geq 1$. That is,
$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}$$
4. Inductive step: (To show $P(k+1)$ is true.)
   4.1. $1 + 2 + \cdots + k + (k+1) = \frac{k(k+1)}{2} + (k+1) = \frac{(k+1)\big((k+1)+1\big)}{2}$ (by basic algebra)
   4.2. Therefore $P(k+1)$ is true.
5. (We have proved $P(1)$ as well as $P(k) \rightarrow P(k+1)$)
   Therefore, $P(n)$ is true for $n \in \mathbb{Z}^+$.

Texts in green here are for explanation; you may omit it in your solution.

We assume $P(k)$ and this is how we make use of $P(k)$.

Mathematical Induction

From the previous slide:

$$\underbrace{1 + 2 + 3 + \cdots + n}_{\text{Open form*}} = \underbrace{\frac{n(n+1)}{2}}_{\text{Closed form}}$$

- An open form describes a quantity in terms of a sequence of operations (often with ellipsis $\cdots$ or large operators such as $\sum$).
- A closed form gives the same quantity directly in terms of n, without the expanding sum (or recurrence).
- The open form shows the pattern; the closed form makes a claim about the pattern; mathematical induction proves the claim.

Note: "Open form" is not a standardized term. $1 + 2 + 3 + \cdots + n$ is usually called summation/expanded form.

Mathematical Induction

Example #9:

Use mathematical induction to prove:

Theorem 5.2.3 (5th: 5.2.2) Sum of a Geometric Sequence

For any real number $r \neq 1$, and any integers $n \geq 0$,

$$\sum_{i=0}^{n} r^i = \frac{r^{n+1}-1}{r-1}$$

Proof (by *mathematical induction*):

1. Let $P(n) \equiv \left(\sum_{i=0}^{n} r^i = \frac{r^{n+1}-1}{r-1}\right), r \neq 1, n \geq 0$. (Set up predicate.)
2. Basis step: $r^0 = 1 = \frac{r^1-1}{r-1}$, therefore $P(0)$ is true.
3. Assume $P(k)$ is true for arbitrary $k \geq 0$. That is, $\sum_{i=0}^{k} r^i = \frac{r^{k+1}-1}{r-1}$
4. Inductive step: (To show $P(k+1)$ is true.)

   4.1. $\sum_{i=0}^{k+1} r^i = \sum_{i=0}^{k} r^i + r^{k+1} = \frac{r^{k+1}-1}{r-1} + r^{k+1} = \frac{r^{k+1}-1+r^{k+1}(r-1)}{r-1} = \frac{r^{((k+1)+1)}-1}{r-1}$

   4.2. Therefore $P(k+1)$ is true.
5. Therefore, $P(n)$ is true for $n \geq 0$.

Mathematical Induction

Example #10:

Use mathematical induction to prove:

Proposition 5.3.1 (5th: 5.3.2)

For all integers $n \geq 0, 2^{2n} - 1$ is divisible by 3.

Proof (by *mathematical induction*):

1. Let $P(n) \equiv (3 \mid (2^{2n} - 1))$ for all integers $n \geq 0$.
2. Basis step: $2^{2 \cdot 0} - 1 = 0$ is divisible by 3, therefore $P(0)$ is true.
3. Assume $P(k)$ is true, that is, $3|(2^{2k} - 1)$ for arbitrary $k \geq 0$.
   3.1 This means that $2^{2k} - 1 = 3r$ for some integer r (by the definition of divisibility).
4. Inductive step: (To show $P(k+1)$ is true.)
   4.1. $2^{2(k+1)} - 1 = 2^{2k} \cdot 4 - 1 = 2^{2k} \cdot (3+1) - 1 = 2^{2k} \cdot 3 + (2^{2k} - 1)$
   $= 2^{2k} \cdot 3 + 3r = 3(2^{2k} + r)$
   4.2. Since $3|(2^{2(k+1)} - 1)$, therefore $P(k+1)$ is true.
5. Therefore, $P(n)$ is true for all integers $n \geq 0$.

Mathematical Induction

Example #11:

Use mathematical induction to prove:

Proposition 5.3.2 (5th: 5.3.3)

For all integers $n \geq 3, 2n + 1 < 2^n$.

Proof (by *mathematical induction*):

1. Let $P(n) \equiv (2n + 1 < 2^n), \forall n \in \mathbb{Z}_{\geq 3}$.
2. Basis step: $2 \cdot 3 + 1 = 7 < 8 = 2^3$, therefore $P(3)$ is true.
3. Assume $P(k)$ is true, that is $2k + 1 < 2^k$ for arbitrary $k \geq 3$.
4. Inductive step: (To show $P(k + 1)$ is true.)
   4.1. $2(k + 1) + 1 = 2k + 3 = (2k + 1) + 2 < 2^k + 2 < 2^k + 2^k = 2^{k+1}$
   (because $2 < 2^k$ for all integers $k \geq 2$)
   4.2. Therefore $P(k + 1)$ is true.
5. Therefore, $P(n)$ is true for all integers $n \geq 3$.

Mathematical Induction

Example #12: A negative example

Claim: All cows have the same colour.



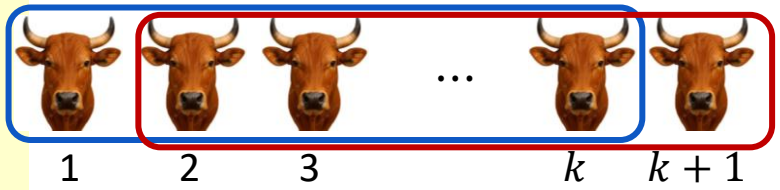
Mathematical Induction

Example #12:

Claim: All cows have the same colour.

Proof (by *mathematical induction*):

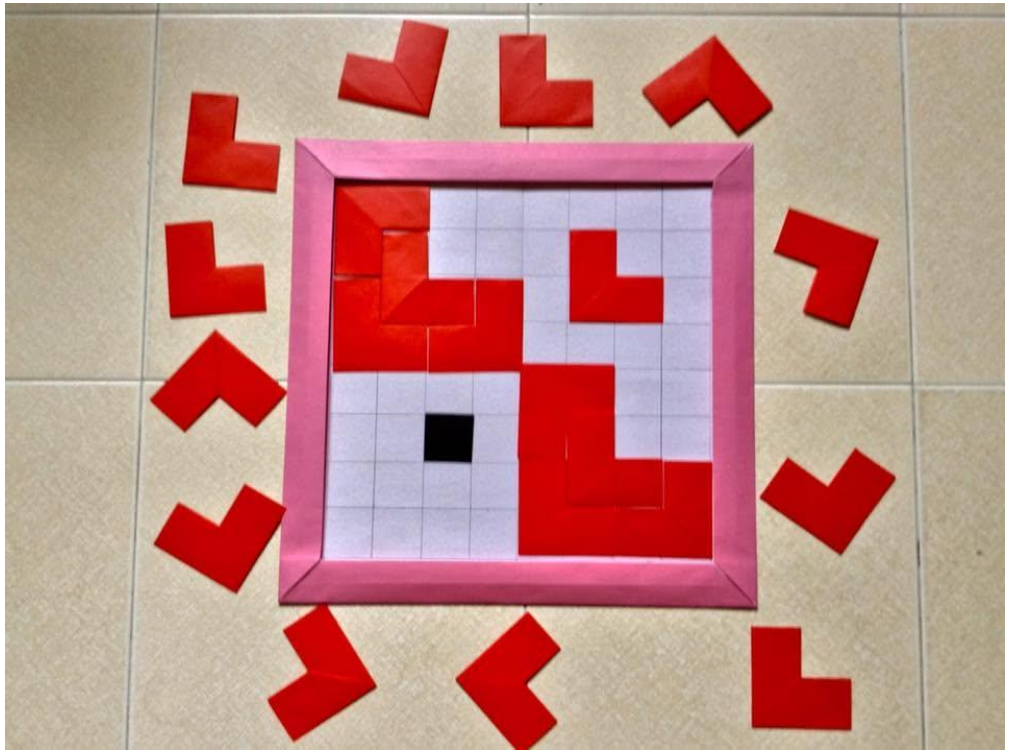
1. Let $P(n) \equiv$ (Any group of n cows have the same colour), $\forall n \in \mathbb{Z}^+$.
2. Basis step: Clearly, a single cow has one colour, so $P(1)$ is true.
3. Assume $P(k)$ is true for arbitrary $k \geq 1$. (A group of k cows have the same colour.)
4. Inductive step: (To show $P(k+1)$ is true.)
   - 4.1. In any group of $k+1$ cows, number them from 1 to $k+1$.
   - 4.2. Then cows #1 to #k form a group of k cows, which have the same colour by the Inductive Hypothesis.
   - 4.3. Similarly, cows #2 to #$k+1$ have the same colour.
   - 4.4. Now, cows #2 to #k are common to both groups, and cows don't change colour!
   - 4.5. Thus cow #$k+1$ has the same colour as cow #1, which means all $(k+1)$ cows have the same colour.
   - 4.6. Therefore, $P(k+1)$ is true.
5. Therefore, $P(n)$ is true, i.e., all cows have the same colour!

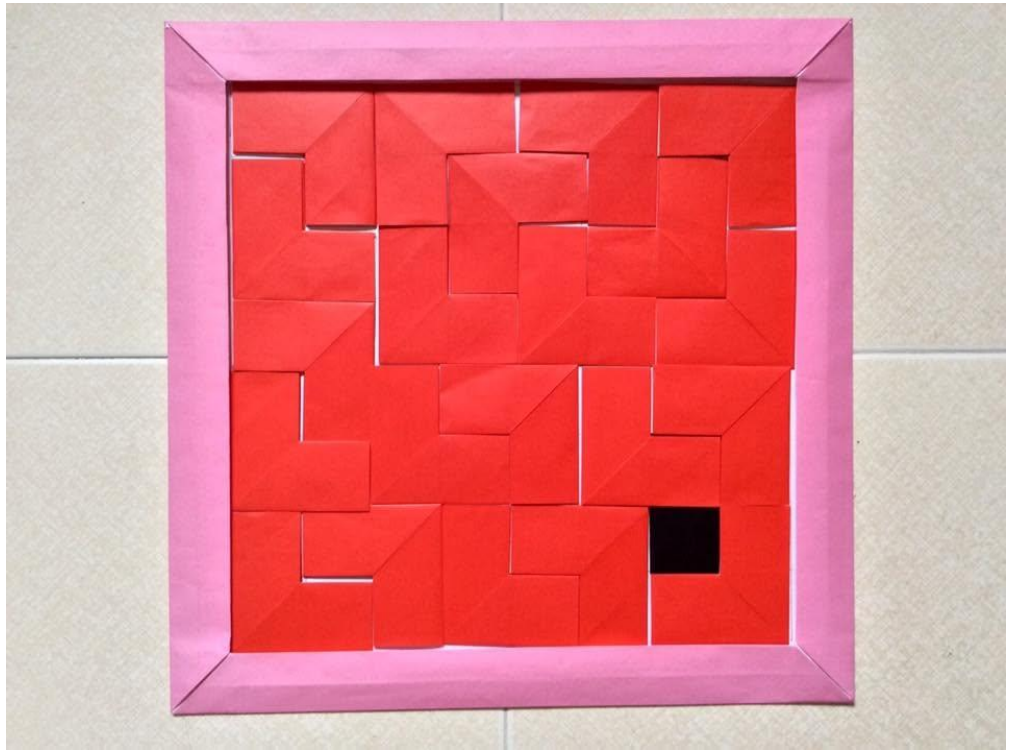


What is the flaw?

Line 2
Line 3
Line 4.1
Line 4.2
Line 4.3
Line 4.4
Line 4.5
Line 4.6

PollEv.com/aarontan





Mathematical Induction

PMI is not restricted to proving formulas.

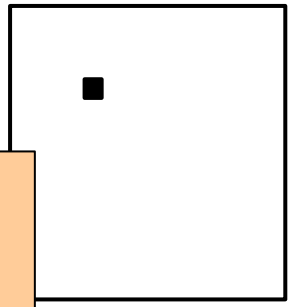
Example #13: For $n \in \mathbb{Z}^+$, any $2^n \times 2^n$ board with one square removed can be tiled by L-trominoes.

Proof (by *mathematical induction*):

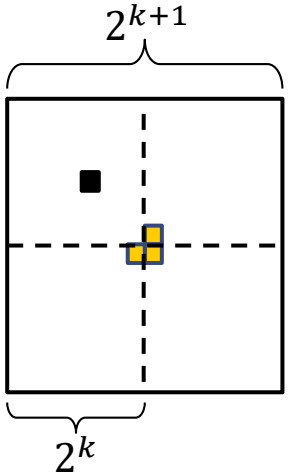
1. Let $P(n) \equiv (2^n \times 2^n$ board with one square removed can be tiled by L-trominoes), $\forall n \in \mathbb{Z}^+$.
2. Basis step: $P(1)$ is true as such a board is an L-tromino.
3. Assume $P(k)$ is true for arbitrary $k \geq 1$.
4. Inductive step: (To show $P(k+1)$ is true.)
   - 4.1. Let B be a $2^{k+1} \times 2^{k+1}$ board with one square removed.
   - 4.2. Divide B into four $2^k \times 2^k$ quadrants.
   - 4.3. Let Q be the quadrant containing the removed square.
   - 4.4. Remove one L-tromino from the centre of B such that each quadrant other than Q has one square removed.
   - 4.5. We have four $2^k \times 2^k$ quadrants, each with one square removed.
   - 4.6. By the induction hypotheses, each quadrant can be tiled by L-trominoes.
   - 4.7. Therefore, $P(k+1)$ is true.
5. Therefore $\forall n \in \mathbb{Z}^+ \; P(n)$ is true.

L-tromino





Golomb's Theorem of Tromino Tiling



Mathematical Induction

AI generated proof: For every positive integer n, $n^2 + n + 1 \leq 3n$.

Base case. For $n = 1$,

$$1^2 + 1 + 1 = 3 \leq 3.$$

Thus $P(1)$ is true.

Inductive hypothesis. Assume that for some positive integer k,

$$k^2 + k + 1 \leq 3k.$$

Inductive step. We need to show that

$$(k+1)^2 + (k+1) + 1 \leq 3(k+1).$$

Starting from the left-hand side,

$$\begin{aligned}(k+1)^2 + (k+1) + 1 &= k^2 + 3k + 3 \\ &= (k^2 + k + 1) + (2k + 2).\end{aligned}$$

By the induction hypothesis,

$$k^2 + k + 1 \leq 3k.$$

Therefore,

$$(k+1)^2 + (k+1) + 1 \leq 3k + (2k + 2).$$

Since $2k + 2 \leq 3$ for positive k,

$$3k + (2k + 2) \leq 3k + 3 = 3(k+1).$$

Hence

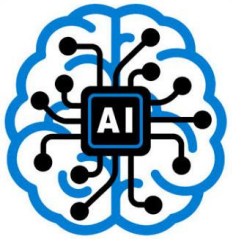
$$(k+1)^2 + (k+1) + 1 \leq 3(k+1).$$

Therefore, by mathematical induction,

$$n^2 + n + 1 \leq 3n$$

for every positive integer n. □

Mathematical Induction



Still AI-generated. To prove: Every integer ≥ 2 is divisible by 2.

Base case: 2 is divisible by 2.

Assume k is divisible by 2. Thus

$$k = 2m$$

for some integer m.

Now consider $k+1$. By the induction hypothesis, $k+1$ is divisible by 2, so

$$k+1 = 2r$$

for some integer r.

Hence $P(k+1)$ is true.

Therefore, by mathematical induction, every positive integer is divisible by 2. □

Common error: the induction hypothesis gives us only $P(k)$.
It does not give us $P(k+1)$.

This proof uses this strategy:

- Assume $P(k)$ is true.
- By the induction hypothesis, $P(k+1)$ is also true.
- Hence $P(k+1)$ follows.
- Therefore, by mathematical induction, $P(n)$ is true for all n.

Mathematical Induction: Common Mistakes (1/3)

1. Assuming what you are trying to proof.
   - The induction hypothesis is $P(k)$, not $P(k+1)$. A proof that effectively says "by induction, $P(k+1)$ is true" is circular reasoning.
2. Failing to use the induction hypothesis at all.
   - Writing a long derivation for $P(k+1)$ but never use $P(k)$. This becomes a direct proof, not proof by induction.
3. Proving the wrong implication.
   - The required step is $P(k) \rightarrow P(k+1)$. Students sometimes prove the converse, $P(k+1) \rightarrow P(k)$, which doesn't establish the induction chain.
4. Treating k as a particular number rather than arbitrary.
   - The inductive step must work for an **arbitrary** k in the required range.
5. Using n instead of k in the inductive step (this is rather commonly seen online)
   - Writing $P(n) \rightarrow P(n+1)$ in the inductive step for proving $P(n)$ for all integers $n \geq 1$.

Mathematical Induction: Common Mistakes (2/3)

6. Assuming $P(k)$ is true for all k.
   - A common wording in the inductive step is "Assume $P(k)$ is true for all $k \geq 1$." But this assumes the entire theorem!
   - It should be something like: "Let $k \geq 1$ be arbitrary and assume $P(k)$ is true."
7. Making an unjustified algebraic/inequality step to force $P(k+1)$.
   - For example, the student reaches $2^{k+1} < 2k^2$ and then simply claims that $2k^2 < (k+1)^2$ without verifying when that inequality is actually true. This is especially dangerous because the induction structure looks perfect.

Induction is like a row of dominoes:
Knocking over the first one is not enough,
Explaining how one knocks over the next is not enough –
you must make sure there isn't a gap somewhere in the middle.

Mathematical Induction II

Theorem 4.3.3 (5th: 4.4.3) Transitivity of Divisibility

For all integers a, b and c, $a \mid b \wedge b \mid c \rightarrow a \mid c$.

Example #14: Any integer > 1 is divisible by a prime number.

Proof (by *mathematical induction*)

1. Let $P(n) \equiv (n$ is divisible by a prime), for $n > 1$.
2. Basis step: $P(2)$ is true since 2 is divisible by 2.
3. Inductive step: To show that for some $k \geq 2$, if $P(k)$ is true then $P(k+1)$ is also true.

   How to proceed???

Say, $14\ (= 2 \times 7)$ is divisible by a prime, how do we show that 15 is divisible by a prime?

How do we relate $k\ (14)$ to $k+1\ (15)$?

Mathematical Induction II

Principle of Strong Mathematical Induction

Let $P(n)$ be a property that is defined for integers n, and let a and b be fixed integers with $a \leq b$. Suppose the following two statements are true:

1. $P(a), P(a+1), \ldots,$ and $P(b)$ are all true. **(basis step)**
2. For any integer $k \geq b$, if $P(i)$ is true for all integers i from a through k, then $P(k+1)$ is true. **(inductive step)**

Then the statement

$$\text{for all integers } n \geq a,\ P(n)$$

is true. (The supposition that $P(i)$ is true for all integers i from a through k is called the **inductive hypothesis.** Another way to state the inductive hypothesis is to say that $P(a), P(a+1), \ldots, P(k)$ are all true.)

Mathematical Induction II

Comparison between "weak" and "strong" induction.

Let $P(n)$ denotes the property on all integers $n \geq a$.

Weak (regular) induction (or 1PI)

If

- $P(a)$ holds
- For every $k \geq a, P(k) \Rightarrow P(k+1)$

Then $P(n)$ holds for all $n \geq a$.

We may prove strong induction from weak and weak induction from strong (proofs omitted).
This means both types of induction are equal in "power".

Strong induction (or 2PI)

If

- $P(a)$ holds
- For every $k \geq a, \big(P(a) \wedge P(a+1) \wedge \cdots \wedge P(k)\big) \Rightarrow P(k+1)$

Then $P(n)$ holds for all $n \geq a$.

Hence, using more neutral terms, we can call the regular/strong versions the First Principle of Mathematical Induction (1PI) and Second Principle of Mathematical Induction (2PI) respectively.

Strong induction (or 2PI) (variation – other variations possible)

If

- $P(a), P(a+1), \ldots, P(b)$ hold
- For every $k \geq a, P(k) \Rightarrow P(k+b-a+1)$

Then $P(n)$ holds for all $n \geq a$.

Mathematical Induction II

Theorem 4.3.3 (5th: 4.4.3) Transitivity of Divisibility

For all integers a, b and c, $a \mid b \wedge b \mid c \rightarrow a \mid c$.

Example #14: Any integer > 1 is divisible by a prime number.

Proof (by *2PI*):

1. Let $P(n) \equiv$ (n is divisible by a prime), for $n > 1$.
2. Basis step: $P(2)$ is true since 2 is divisible by 2.
3. Inductive step: To show that for arbitrary $k \geq 2$, if $P(i)$ is true for all integers i from 2 through k, then $P(k+1)$ is also true.
   - 3.1. Case 1 ($k+1$ is prime): In this case $k+1$ is divisible by a prime number which is itself.
   - 3.2. Case 2 ($k+1$ is not prime): In this case $k+1 = ab$ where a and b are integers with $1 < a < k+1$ and $1 < b < k+1$.
     - 3.2.1. Thus, in particular, $2 \leq a \leq k$ and so by inductive hypothesis, a is divisible by a prime number p.
     - 3.2.2. In addition, because $k+1 = ab$, so $k+1$ is divisible by a.
     - 3.2.3. By transitivity of divisibility, $k+1$ is divisible by a prime p.
4. Therefore any integer greater than 1 is divisible by a prime.

Mathematical Induction II

Example #15: Use 1PI to prove: any whole amount of $\geq \$12$ can be formed by a combination of \$4 and \$5 coins.

Proof (by *1PI*):

1. Let $P(n) \equiv$ (the amount of $\$n$ can be formed by \$4 and \$5 coins) for $n \geq 12$.
2. Basis step: $12 = 3 \times 4$, so three \$4 can be used. Therefore $P(12)$ is true.
3. Assume $P(k)$ is true for arbitrary $k \geq 12$.
4. Inductive step: (To show $P(k+1)$ is true.)
   - 4.1. Case 1: If a \$4 coin is used for $\$k$ amount, replace it by a \$5 coin to make $\$(k+1)$.
   - 4.2. Case 2: If no \$4 coin is used for $\$k$ amount, then $k \geq 15$, so there must be at least three \$5 coins. We can then replace three \$5 coins with four \$4 coins to make $\$(k+1)$.
   - 4.3. In both cases, $P(k+1)$ is true.
5. Therefore, $P(n)$ is true for $n \geq 12$.

Mathematical Induction II

This is the same problem as Example #15.

Example #16: Use 2PI to prove: For all integers $n \geq 12, n = 4a + 5b$ for some $a, b \in \mathbb{N}$.

Proof (by *2PI*):

1. Let $P(n) \equiv (n = 4a + 5b)$, for some $a, b \in \mathbb{N}, n \geq 12$.
2. Basis step: Show that $P(12), P(13), P(14), P(15)$ hold.
   $12 = 4 \cdot 3 + 5 \cdot 0;\ 13 = 4 \cdot 2 + 5 \cdot 1;\ 14 = 4 \cdot 1 + 5 \cdot 2;\ 15 = 4 \cdot 0 + 5 \cdot 3$
3. Assume $P(i)$ holds for $12 \leq i \leq k$ for arbitrary $k \geq 15$.
4. Inductive step: (To show $P(k+1)$ is true.)
   4.1. $P(k-3)$ holds (by induction hypothesis),
   so, $k - 3 = 4a + 5b$ for some $a, b \in \mathbb{N}$
   4.2. $k + 1 = (k - 3) + 4 = (4a + 5b) + 4 = 4(a + 1) + 5b$
   4.3. Hence, $P(k+1)$ is true.
5. Therefore, $P(n)$ is true for $n \geq 12$.

What if we only prove $P(12)$ in the basis step?

Mathematical Induction: Common Mistakes (3/3)

8. Using more than the induction hypothesis provides.
   - In weak induction (1PI), assuming $P(k)$does **not** permit you to use $P(k-1)$, $P(k-2)$, etc. If earlier cases are needed, strong induction (2PI) may be the appropriate formulation.
9. Too few base cases for strong induction (2PI)
   - Suppose you are to claim $P(n)$ for all integers $n \geq 1$. You establish $P(k) \Rightarrow P(k+2)$ but provide only $P(1)$ in the basis step. This only establish $P(1), P(3), P(5)$, … (odd values of n) and not $P(2), P(4), P(6)$, … (even values of n).

Past-Year's Exam Question!

AY2022/23 Sem1

21. Let $f_0, f_1, f_2, \cdots$ be the Fibonacci sequence 0,1,1,2,3,5,8,13,21,…

Prove by mathematical induction that, for all $n \in \mathbb{N}$,

$$\sum_{i=0}^{n} f_i^2 = f_n f_{n+1}.$$

The first line of the proof is given.

1. For each $n \in \mathbb{N}$, let $P(n)$ be the proposition $\sum_{i=0}^{n} f_i^2 = f_n f_{n+1}$.
2. (Basis step) When $n = 0$, $f_0^2 = 0^2 = 0 = 0 \cdot 1 = f_0 f_1$. So $P(0)$ is true.
3. (Induction step)
   - 3.1. Let $k \in \mathbb{N}$ such that $P(k)$ is true, i.e. $\sum_{i=0}^{k} f_i^2 = f_k f_{k+1}$.
   - 3.2. Then $\sum_{i=0}^{k+1} f_i^2 = \sum_{i=0}^{k} f_i^2 + f_{k+1}^2 = f_k f_{k+1} + f_{k+1}^2$. (by the induction hypothesis)
     $= f_{k+1}(f_k + f_{k+1})$ (by basic algebra)
     $= f_{k+1} f_{k+2}$ (by the definition of Fibonacci)
   - 3.3. Thus $P(k+1)$ is true.
4. Therefore, $\forall\, n \in \mathbb{N}\ P(n)$ is true by mathematical Induction.

Well-Ordering Principle (WOP)

Well-Ordering Principle for the Integers

Every nonempty subset of $\mathbb{Z}_{\geq 0}$ has a smallest element.

Proof (by *contradiction*):

1. Suppose not, i.e. let $S \subseteq \mathbb{Z}_{\geq 0}$ be non-empty with no smallest element.
2. For each $n \in \mathbb{Z}_{\geq 0}$, let $P(n)$ be the proposition "$n \notin S$".
3. Inductive step:
   - 3.1. Let $k \in \mathbb{Z}_{\geq 0}$ such that $P(0), P(1), \cdots, P(k-1)$ are true, i.e., $0,1, \cdots, k-1 \notin S$.
   - 3.2. If $k \in S$, then k is the smallest element of S by the induction hypothesis as $S \subseteq \mathbb{Z}_{\geq 0}$, which contradicts our assumption that S has no smallest element
   - 3.3. So $k \notin S$ and thus $P(k)$ is true.
4. Hence $\forall n \in \mathbb{Z}_{\geq 0}\ P(n)$ is true by 2PI.
5. This implies $S = \emptyset$, contradicting line 1 that S is non-empty.

Well-Ordering Principle (WOP)

Well-Ordering Principle for the Integers

Every nonempty subset of $\mathbb{Z}_{\geq 0}$ has a smallest element.

Alternative proof:

Proof (by *contradiction*):

1. Suppose not, i.e. let $S \subseteq \mathbb{Z}_{\geq 0}$ be non-empty with no smallest element.
2. Choose some $a \in S$.
   - 2.1. Since a is not the smallest element, there must be some $a_1 \in S$ with $a_1 < a$.
   - 2.2. But a_1 is not smallest either, so there is $a_2 \in S$ with $a_2 < a_1$.
   - 2.3. Continuing, we get an infinite descending sequence $a > a_1 > a_2 > \cdots$ which cannot occur in $\mathbb{Z}_{\geq 0}$.
4. Hence S must have a smallest element.

Mathematical induction and Well-Ordering Principle are equivalent. (Each can be used to prove the other.)

Well-Ordering Principle (WOP)

Well-Ordering Principle for the Integers

Every nonempty subset of $\mathbb{Z}_{\geq 0}$ has a smallest element.

Deriving induction from WOP:

1. Suppose $P(0)$ is true and $P(k) \Rightarrow P(k+1)$ for all $k \geq 0$.
2. Assume some $P(k)$ is false.
   - 2.1. Consider the set of counterexamples $C = \{k \in \mathbb{Z}_{\geq 0} : P(k) \text{ is false}\}$.
   - 2.2. If $C \neq \emptyset$, WOP says C has a smallest element, say m. (So $P(m)$ is false and m is the smallest number for which the property is false.)
   - 2.3. Since $P(0)$ is true, $m > 0$.
   - 2.4. By the minimality of m, $P(m-1)$ must be true.
   - 2.5. But the inductive step then gives $P(m-1) \Rightarrow P(m)$ which means $P(m)$ is true – a contradiction.
4. Hence induction is derived.

Well-Ordering Principle (WOP)

Deriving induction from WOP:

Well-Ordering Principle for the Integers

Every nonempty subset of $\mathbb{Z}_{\geq 0}$ has a smallest element.

1. Suppose $P(0)$ is true and $P(k) \Rightarrow P(k+1)$ for all $k \geq 0$.
2. Assume some $P(k)$ is false.
   - 2.1. Consider the set of counterexamples $C = \{k \in \mathbb{Z}_{\geq 0} : P(k) \text{ is false}\}$.
   - 2.2. If $C \neq \emptyset$, WOP says C has a smallest element, say m. (So $P(m)$ is false and m is the smallest number for which the property is false.)
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   - 2.4. By the minimality of m, $P(m-1)$ must be true.
   - 2.5. But the inductive step then gives $P(m-1) \Rightarrow P(m)$ which means $P(m)$ is true – a contradiction.
4. Hence induction is derived.



The domino analogy.

Suppose there is at least one domino that doesn't fall. WOP says:

Then there must be a first domino that doesn't fall.

Call it domino m. It can't be domino 1, because we know domino 1 falls (basis step).
So, there is a domino m (>1), the first one, that doesn't fall.
So, domino $m-1$ does fall.
But induction step establishes that: Whenever a domino falls, the next domino falls.
Therefore, if $m-1$ does fall, then m must fall – Contradiction!

Recurrence Relations

Definition

A **recurrence relation** for a sequence $a_0, a_1, a_2, \cdots$ is a formula that relates each term a_k to certain of its predecessors $a_{k-1}, a_{k-2}, \cdots, a_{k-i}$, where i is an integer with $k - i \geq 0$.

If i is a fixed integer , the **initial conditions** for such a recurrent relation specify the values of $a_0, a_1, a_2, \cdots, a_{i-1}$.

If i depends on k, the initial conditions specify the values of $a_0, a_1, a_2, \cdots, a_m$, where m is an integer with $m \geq 0$.

Example #18: Recurrence relation for Fibonacci sequence F_n.

$$F_0 = 0, F_1 = 1$$
$$F_n = F_{n-1} + F_{n-2}, \text{ for } n > 1$$

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, …

Sometimes, we call it a recursive definition.

Examples:

- Recursive definition of *factorial*:

  $0! = 1$

  $n! = n \cdot (n-1)!$ for $n \geq 1$

- Recursive definition of *power*:

  $a^0 = 1$

  $a^n = a^{n-1} \cdot a$ for $n \geq 1$

Recurrence Relations

Example #19: Prove that for any positive integer n, if $a_1, a_2, \cdots, a_n$ and $b_1, b_2, \cdots, b_n$ are real numbers, then $\sum_{i=1}^{n}(a_i + b_i) = \sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i$.

Proof (by *mathematical induction*):

1. Let $P(n) \equiv (\sum_{i=1}^{n}(a_i + b_i) = \sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i)$, for $n \geq 1$.
2. Basis step: $P(1)$ is true since

   $\sum_{i=1}^{1}(a_i + b_i) = a_1 + b_1 = \sum_{i=1}^{1} a_i + \sum_{i=1}^{1} b_i$.
3. Inductive hypothesis: for arbitrary $k \geq 1$,

   $\sum_{i=1}^{k}(a_i + b_i) = \sum_{i=1}^{k} a_i + \sum_{i=1}^{k} b_i$.
4. Inductive step:

$$\begin{aligned}\sum_{i=1}^{k+1}(a_i + b_i) &= \left(\sum_{i=1}^{k}(a_i + b_i)\right) + (a_{k+1} + b_{k+1}) && \text{(by definition of } \textstyle\sum\text{)}\\ &= \sum_{i=1}^{k} a_i + \sum_{i=1}^{k} b_i + (a_{k+1} + b_{k+1}) && \text{(by inductive hypothesis)}\\ &= \cdots\end{aligned}$$

Recurrence Relations

Example #19: Prove that for any positive integer n, if $a_1, a_2, \cdots, a_n$ and $b_1, b_2, \cdots, b_n$ are real numbers, then $\sum_{i=1}^{n}(a_i + b_i) = \sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i$.

4. Inductive step:

$\sum_{i=1}^{k+1}(a_i + b_i) = \left(\sum_{i=1}^{k}(a_i + b_i)\right) + (a_{k+1} + b_{k+1})$ (by definition of $\sum$)

$= \sum_{i=1}^{k} a_i + \sum_{i=1}^{k} b_i + (a_{k+1} + b_{k+1})$ (by inductive hypothesis)

$= \sum_{i=1}^{k} a_i + a_{k+1} + \sum_{i=1}^{k} b_i + b_{k+1}$ (by the associative and commutative laws of algebra)

$= \sum_{i=1}^{k+1} a_i + \sum_{i=1}^{k+1} b_i$ (by definition of $\sum$)

Therefore $P(k+1)$ is true.

5. Therefore $P(n)$ is true for any positive integer n.

Recursively Defined Sets

Definition

Let S be a finite set with at least one element. A **string over** S is a finite sequence of elements from S. The elements of S are called **characters** of the string, and the **length** of a string is the number of characters it contains. The **null string over** S is defined to be the "string" with no characters. It is usually denoted ϵ and is said to have length 0.

Recall in Lecture 7:

Definition: String

Let A be a set. A **string** or a word over A is an expression of the form $a_0 a_1 a_2 \cdots a_{l-1}$ where $l \in \mathbb{Z}_{\geq 0}$ and $a_0, a_1, a_2, \cdots, a_{l-1} \in A$.

Here l is called the **length** of the string. The **empty string** ε is the string of length 0.

Let A^* denote the set of all strings over A.

Recursively Defined Sets

Example #21: Certain configurations of parentheses in algebraic expressions are legal [such as (())() and ()()()], whereas others are not [such as (()))) and ())()(].

Here is a recursive definition to generate the set P of legal configurations of parentheses.

I. Base: $()$ is in P.

II. Recursion:
   a. If E is in P, so is (E).
   b. If E and F are in P, so is EF.

III. Restriction: No configurations of parentheses are in P other than those derived from 1 and 2 above.

Derive the fact that $(())()$ is in P.

Recursively Defined Sets

Example #21: Derive the fact that $(())()$ is in P.

I. Base: $()$ is in P.

II. Recursion:
   a. If E is in P, so is (E).
   b. If E and F are in P, so is EF.

III. Restriction: No configurations of parentheses are in P other than those derived from 1 and 2 above.

1. By I, $()$ is in P.
2. By (1) and IIa, $(())$ is in P [let $E = ()$].
3. By (2), (1) and IIb, $(())()$ is in P [let $E = (())$and $F = ()$].

Recursively Defined Sets

Example #22: Recursive definition of $\mathbb{Z}_{\geq 0}$.

$\mathbb{Z}_{\geq 0}$ is the unique set with the following properties:

(1. what the founders are) $0 \in \mathbb{Z}_{\geq 0}$. (base clause)

(2. what the constructors are) If $x \in \mathbb{Z}_{\geq 0}$, then $x + 1 \in \mathbb{Z}_{\geq 0}$. (recursion clause)

(3. nothing more) Membership for $\mathbb{Z}_{\geq 0}$ can always be demonstrated by (finitely many) successive applications of the clauses above. (minimality clause)

$\mathbb{Z}_{\geq 0}$

$0 \underset{+1}{} \quad 1 \underset{+1}{} \quad 2 \underset{+1}{} \quad 3 \underset{+1}{} \quad 4 \underset{+1}{} \quad \cdots$

Recursively Defined Sets

Example #23: Recursive definition of $2\mathbb{Z}$ (the set of even integers).

$2\mathbb{Z}$ is the unique set with the following properties:

(1. what the founders are) $0 \in 2\mathbb{Z}$. (base clause)

(2. what the constructors are) If $x \in 2\mathbb{Z}$, then $x - 2, x + 2 \in 2\mathbb{Z}$. (recursion clause)

(3. nothing more) Membership for $2\mathbb{Z}$ can always be demonstrated by (finitely many) successive applications of the clauses above. (minimality clause)

$2\mathbb{Z}$

$$\cdots \quad \underset{-2}{-4} \quad \underset{-2}{-2} \quad \underset{-2}{0}_{+2} \quad 2_{+2} \quad 4_{+2} \quad \cdots$$

Structural Induction

Recursive definition of of a set S.

(base clause) Specify that certain elements, called founders, are in S:
if c is a founder, then $c \in S$.

(recursion clause) Specify certain functions, called constructors, under which the set S is closed: if f is a constructor and $x \in S$, then $f(x) \in S$.

(minimality clause) Membership for S can always be demonstrated by (finitely many) successive applications of the clauses above.

Structural induction over S.

To prove that $\forall x \in S\ P(x)$ is true, where each $P(x)$ is a proposition, it suffices to:

(basis step) show that $P(c)$ is true for every founder c; and

(induction step) show that $\forall x \in S \left(P(x) \Rightarrow P\big(f(x)\big)\right)$ is true for every constructor f.

In other words, if all the founders satisfy a property P, and P is preserved by all constructors, then all elements of S satisfy P.

Structural Induction

Example #24: Define a set S recursively as follows:

(1) $1 \in S$. (base clause)

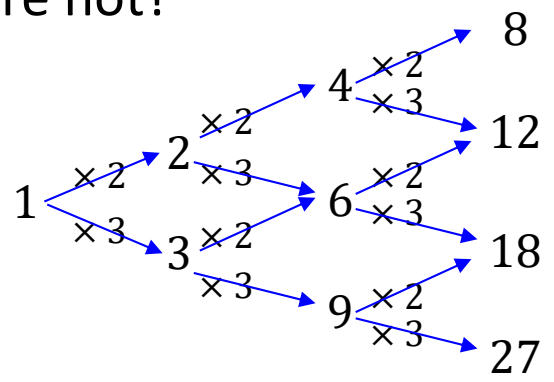
(2) If $x \in S$, then $2x \in S$ and $3x \in S$. (recursion clause)

(3) Membership of S can always be demonstrated by (finitely many) successive applications of the clauses above. (minimality clause)

Which of the numbers 9,10,11,12,13 are in S? Which are not?

$9,12 \in S$

$10,11,13 \notin S$



Structural induction over S:

To prove that $\forall x \in S\ P(n)$ is true, where each $P(n)$ is a proposition, it suffices to:

(basis step): show that P(1) is true; and

(induction step): show that $\forall x \in S\big(P(x) \Rightarrow P(2x) \wedge P(3x)\big)$ is true.

Past-Year's Exam Question!

AY2020/21 Sem2

11. Recall that $\{s, u\}^*$ denotes the set of all strings over $\{s, u\}$.

Define a subset $T \subseteq \{s, u\}^*$ recursively as follows.,

- $su \in T$. (base clause)
- If $\sigma, \tau \in T$, then $s\sigma u \in T$ and $s\sigma\tau u \in T$. (recursion clause)
- Membership for T can always be demonstrated by (finitely many) successive applications of clauses above. (minimality clause)

Which of the following is/are in T?

A. ssssusuuusuu

$su \rightarrow ssusuu \rightarrow sssusuuu \rightarrow ssssusuuusuu$

B. ssussusususuuu

C. ssussuusssuu

D. usususussuu

One can show the following statements by structural induction over T.

(i) All elements of T start with s and end with u.

(ii) All elements of T of length at least 3 start with ss and end with uu.

(iii) In no element of T can there be an occurrence of usus.

(iv) In every element of T, the number of s's is equal to the number of u's.

We can use (iii) to eliminate B, use (iv) to eliminate C, and use (i) to eliminate D.

Past-Year's Exam Question!

AY2022/23 Sem1

14. Define a set S recursively as follows:
    (1) $1 \in S$ and $2 \in S$. (base clause)
    (2) If $x, y \in S$, then $x/y \in S$. (recursion clause)
    (3) Membership of S can always be demonstrated by (finitely many) successive applications of the clauses above. (minimality clause)

Which of the following statements is true?

A. $\forall x \in S\ (x \in \mathbb{Z} \Rightarrow x \text{ is even})$.

B. $\forall x \in S\ (x \in \mathbb{Q} \Rightarrow x < 1)$.

C. $\forall x, y \in S\ (x + y \geq 1)$.

D. $|S| = |\mathbb{Z}|$.

E. None of the above.

The elements of S are the integers powers of 2, i.e. $S = \{2^n : n \in \mathbb{Z}\}$ or $S = \{\cdots, \frac{1}{8}, \frac{1}{4}, \frac{1}{2}, 1, 2, 4, 8, \cdots\}$.

(A) is false. Counterexample: 1
(B) is false. Counterexample: 2
(C) is false. Counterexample: $\frac{1}{2} + \frac{1}{4} = \frac{3}{4} < 1$.
(D) We can define a bijection $f : \mathbb{Z} \to S$ as $f(n) = 2^n$.

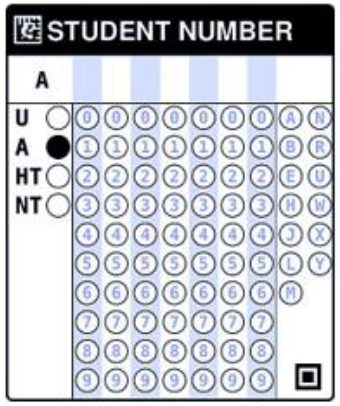
How to shade your Student Number in the Midterm Test (and Exam)

Midterm Test (CS1231S Discrete Structures)

Time allowed: 1 hour 30 minutes

INSTRUCTIONS

1. Write your **Student Number** on the right AND, using a **pencil** (2B or above), shade the corresponding circle **completely** in the grid for each digit or letter. DO NOT WRITE YOUR NAME!
2. Zero mark will be given if you write/shade your Student Number incompletely or incorrectly.
3. There are **TWO (2) pages** in this Answer Sheet.
4. All answers must be written in the boxes provided.
5. You must submit only this **ANSWER SHEET** and no other documents. Do not attach any other sheet.
6. An excerpt of the question may be provided to aid you in answering in the correct box. It is not the exact question. You should still refer to the original question in the question paper.
7. You may write your answers in pencil (2B or above) or pen. Pencil is preferred in case you need to erase and rewrite your answers.
8. The maximum mark for this paper is 50.
9. **Marks may be deducted** for (i) illegible handwriting, and/or (ii) excessively long answer.
10. Each multiple choice question is intended to have only one answer. Please shade the appropriate bubble in **pencil**.



For Examiner's Use Only		
Question	Total	Obtained
Q1-Q12	24	
Q13	5	
Q14	3	
Q15	6	
Q16	6	
Q17	4	
Q18	2	
Total	**50**	

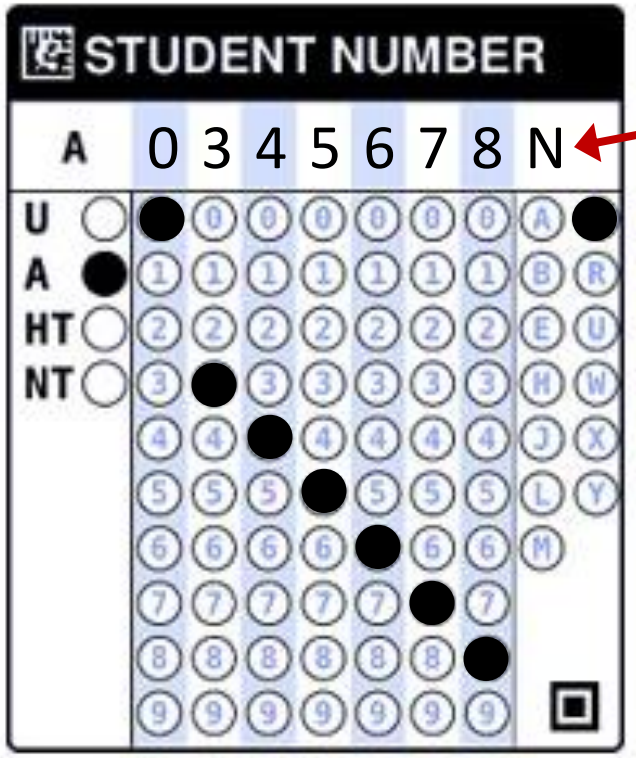
Part A: Multiple Choice Questions (Total: 24 marks)

Please shade clearly only ONE bubble for each question using **pencil**.

	(A)	(B)	(C)	(D)	(E)
1.	○	○	○	○	○
2.	○	○	○	○	○
3.	○	○	○	○	○
4.	○	○	○	○	○
5.	○	○	○	○	○
6.	○	○	○	○	○

	(A)	(B)	(C)	(D)	(E)
7.	○	○	○	○	○
8.	○	○	○	○	○
9.	○	○	○	○	○
10.	○	○	○	○	○
11.	○	○	○	○	○
12.	○	○	○	○	○

- Bring the following:
  - Student Card (or any Photo-ID if you do not have your student card)
  - 2B (or above) pencils and eraser
  - Own blank papers for rough work



- Write your Student Number in pencil.
- Shade the corresponding bubbles for your Student Number in pencil.
- Check that you have written/shaded your Student Number correctly.
- Double check again.
- Do not write your name anywhere.
- Your identity is determined by your shaded Student Number. If you do not shade it correctly, we are unable to identify you!

- Do not shade too lightly.

- Do not cross/cancel out a wrong answer. Erase it completely.

Part A: Multiple Choice Questions (Total: 24 marks)

Please shade clearly only ONE bubble for each question using **pencil**.

	(A)	(B)	(C)	(D)	(E)
1.	○	◌	○	○	○
2.	○	○	○	○	○
3.	○	○	○	○	○
4.	○	○	○	○	○
5.	○	○	○	○	○
6.	○	○	○	○	○

	(A)	(B)	(C)	(D)	(E)
7.	○	○	●	○	●
8.	○	○	○	○	○
9.	○	○	○	○	○
10.	○	○	○	○	○
11.	○	○	○	○	○
12.	○	○	●	◌	○

- Erase your wrong answer completely, or the software may still pick up two answers.

Next Friday, 8 October 2026, is
So, no lecture!



END OF FILE

SoC Student Life & AI Survey

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Prefer to use your laptop?

tinyurl.com/PerceptionAI-SL
tinyurl.com/58r5nwdm

Thank you for your participation!



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